Randomized Complete Block Designs (and Some Practical Side Topics)

PUBH 526

Design and Analysis of Randomized Trials in Public Health

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Side Topic: Participant IDs

Careful about maintaining privacy!

- Avoid typing whenever possible
  - Drop down menus on electronic forms/data entry; preprinted stickers for paper forms or sample collection; etc.
- Other descriptive information for verification
- IDs can be entered twice
- Useful to have a fixed number of digits – if a digit is missing, you notice immediately
- Useful to use certain digits to indicate specific study sites or other features of your sample (example from Etheredge et al.)
- Check digits (example from Etheredge et al.)
- Check for duplicate IDs in your datasets

Randomization

- “randomization 1.sas”
- Excel: “=RAND()”

Block Randomization "Design" Stage, not Analysis Stage.

- Divide your participants into blocks, then randomize within those blocks
  - For example, if you have blocks of 20 participants and 2 treatments, for each set of 20 participants, you'll have 10 in treatment 1 and 10 in treatment 2
- Treatments will then be close to balanced at any given time
- Creates balance across temporal trends or changes in implementation; may be useful if study stopped early
- Analysis:
  - RCTs often use block randomization but ignore the blocks in the analysis ☹
  - Including blocks in the analysis may increase power (but also use up degrees of freedom, which can decrease power)
- Friedman et al. Chapter 6

Randomized Complete Block Design (RCBD)

- Friedman et al. Chapter 6
- Stratified randomization
- What does it mean to stratify?
- You can stratify at randomization or during analysis (post-stratification, ANCOVA)
  - Depends on your objectives and hypotheses
  - Which factors do you stratify by?
  - How influential are these factors on your outcomes?
  - Which hypotheses do you want to be adequately powered?

Randomized Complete Block Design

- "blocking.sas"
- Major effect on the order of study activities!!
  - (you should have a protocol and manual of operations ahead of time!)

RCBD

- Blocking factors should be important (per your theory)
- "Nuisance factors" - affects the response, but are not the primary question
- In public health applications, we may be interested in:
  - Effect modification
  - Equitability
  - Geographical/contextual differences
- Before a study, identify potential factors
  - What is their prevalence in the population?
  - Do you want to make inferences about subgroups?
  - Do you have the power/resources to do so?

RCBD

- Unknown factors
  - Randomization will help to distribute these factors across your treatments/interventions
    - Protection from bias
  - The effects of these factors on your outcomes all go into ε
    - Inflate your error variance, create noise, reduce power
- Known but uncontrollable factors
  - Post-stratification, ANCOVA
- Known and controllable factors
  - Have the option of a block design
- Special case: Paired t-test situations (each pair is a block)

Statistical Model

We have a treatments and b blocks

Same stats/math as Lecture 9,
but different interpretation.

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk} \begin{cases} i = 1, 2, \dots, a \\ j = 1, 2, \dots, b \\ k = 1, 2, \dots, n_{ij} \end{cases}$$

$$\varepsilon_{ijk} \sim iid\ N(0, \sigma^2)$$

- τ_i is effect of treatment *i* (ignoring block)
- β_j is effect of block *j* (ignoring treatment)
- $(\tau\beta)_{ij}$ is the interaction effect of treatment *i* and block *j*

Lecture 9 is EM ':j"

Model Estimates

- Under the "sum to zero" parameter constraints:

$$\widehat{\mu} = \overline{y}_{...}$$

$$\widehat{\tau}_i = \overline{y}_{i..} - \overline{y}_{...}$$

$$\widehat{\beta}_j = \overline{y}_{.j.} - \overline{y}_{...}$$

$$\widehat{(\tau\beta)}_{ij} = \overline{y}_{ij.} - \overline{y}_{i..} - \overline{y}_{.j.} + \overline{y}_{...}$$

- The predicted value is ij combination average, so

$$\widehat{y}_{ijk} = \overline{y}_{ij.} \text{ and } \hat{\epsilon}_{ijk} = y_{ijk} - \overline{y}_{ij.}$$

- We would get the same predicted values and residuals if we combined factors into single treatment factor and used one-way ANOVA

Partitioning the Sum of Squares

- Rewrite each observation as

$$y_{ijk} = \overline{y}_{...} + (\overline{y}_{i..} - \overline{y}_{...}) + (\overline{y}_{.j.} - \overline{y}_{...}) + (\overline{y}_{ij.} - \overline{y}_{i..} - \overline{y}_{.j.} + \overline{y}_{...}) + (y_{ijk} - \overline{y}_{ij.})$$

- The total SS $\sum (y_{ijk} - \overline{y}_{...})^2$ can be broken up into

$$bn \sum_i (\overline{y}_{i..} - \overline{y}_{...})^2 +$$
$$an \sum_j (\overline{y}_{.j.} - \overline{y}_{...})^2 +$$
$$n \sum_i \sum_j (\overline{y}_{ij.} - \overline{y}_{i.} - \overline{y}_{.j} + \overline{y}_{..})^2 + \dots$$

- $\mathrm{SS_A} + \mathrm{SS_B} + \mathrm{SS_{AB}} + \mathrm{SS_E}$
- Under normality, all SS/σ^2 independent

ANOVA Table

Source of variation	Sum of squares	Degrees of freedom	Mean square	F
Treatment	SS_τ	a-1	$MSA = SS_\tau/(a-1)$	MSA/MSE
Block	SS_β	b-1	$MSB = SS_\beta/(b-1)$	MSB/MSE
Treatment x block interaction	$SS_{\tau\beta}$	(a-1)(b-1)	$MSAB = SS_{\tau\beta}/((a-1)(b-1))$	MSAB/MSE
Error	SS_E	$\sum\sum n_{ij}$-ab	$MSE = SS_E/(\sum\sum n_{ij}-ab)$	
Total	SS_T	$\sum\sum n_{ij}$-1		

- Look at familiar? (recall effect modification) Same stats/math as Lecture 9, but different interpretation.
- What happens when you have 1 observation per treatment*block combination?

Hypothesis Testing

- If the same number of observations in each treatment*block combination:

$$\mathrm{E}(\mathrm{MS_E}) = \sigma^2$$
$$\mathrm{E}(\mathrm{MS_A}) = \sigma^2 + bn\sum \tau_i^2/(a-1)$$
$$\mathrm{E}(\mathrm{MS_B}) = \sigma^2 + an\sum \beta_j^2/(b-1)$$
$$\mathrm{E}(\mathrm{MS_{AB}}) = \sigma^2 + n\sum (\tau\beta)_{ij}^2/(a-1)(b-1)$$

- Use F-tests to test equality of A, B, and AB effects

Caution on Interaction Effects

- Based on the linear model:

$$\begin{aligned} \overline{y}_{i..} &= \mu + \tau_i + \overline{\beta}_{.} + \overline{(\tau\beta)}_{i.} \\ \overline{y}_{i..} - \overline{y}_{i'..} &= \tau_i - \tau_{i'} + \overline{(\tau\beta)}_{i.} - \overline{(\tau\beta)}_{i'.} \end{aligned}$$

- Comparison of treatment effects depends on the corresponding interaction effects

Caution on Interaction Effects

- In RCBD, include main effects and interaction effects in your model
  - Test for interaction effects first
    - If interaction effects are not significant, it makes sense to talk about main effects
    - If interaction effects are significant, the effect of treatments vary depending on the blocking factor – so you can't talk about the effect of a treatment in general
  - I keep interaction effects in the model even if not significant, so that I can show that non-significance to my audience

“foodsecurity.sas”

- Stratified randomization
- 3 agroecological zones (AEZ): high, medium, and low altitude
- Intervention (promotion of home gardening) vs. control (no intervention)
- Household food insecurity access scale (HFIAS) score: 0 (food secure) – 27 (very food insecure)
  - Lower number is better
- Descriptive stats:
  - Where do you expect balance?
  - Caution about inference: Proportion of people in each stratum in your sample differs from proportion in each stratum in the population